Quantifying Investor Sentiment with Multimodal Data in the Chinese Stock Market

By Xiaobin Liu and Yi Yu*

This study is based on the visual and textual information from the homepage of Sina News. It employs the Vision Transformer model to classify the sentiment of news images and constructs a photo-based pessimism index, PhotoPes. Additionally, a pre-trained language model based on BERT is used to build a text-based pessimism index, TextPes. Empirical analysis over the 2014–2026 sample period reveals a robust negative association between lagged news-image sentiment and subsequent market returns, with $PhotoPes_{t-3}$ *delivering negative coefficients across multiple Chinese stock indices—particularly the ChiNext Composite (in-sample* $|t| = 2.81$*, raw* $p \approx 0.005$*, suggestive after Bonferroni correction across the* 5×5 *index–lag family) and the mid/small-cap SZSE Composite and CSI 500 (in-sample 10% level). An out-of-sample exercise using a pre-OOS-cross-validated Ridge model yields a small but statistically detectable separation from the historical-mean forecast for the ChiNext (one-sided* $R^2_{OOS} = 0.015\%$*,* $p = 0.042$*). Image and text sentiment indicators provide incremental and largely independent predictive content for stock market returns, supporting the value of combining modalities rather than relying on either source alone. The findings should be interpreted as evidence on the existence of a small visual-sentiment predictability channel, not as a quantitatively large or tradable effect.*

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In recent years, investor sentiment, as a significant research direction in behavioral finance, has garnered considerable attention from the academic community due to its widespread impact on market fluctuations and investor behavior. Tetlock (2007) argued that investor sentiment not only directly affects market returns and trading volumes but also has a significant effect on market volatility, especially during periods of market panic or intense fluctuation. Quantifying investor sentiment is a key focus in financial research. In existing studies, scholars have mostly focused on quantifying the sentiment of textual content from sources like news articles and social media data, using Natural Language Processing (NLP)

* Liu and Yu contributed equally to this work. Author order is alphabetical. Liu: Associate Professor, Lingnan College, Sun Yat-sen University (liuxb53@mail.sysu.edu.cn). Yu: M.S. in Data Science student, New York University (yy5919@nyu.edu); B.A. in Economics, Lingnan College, Sun Yat-sen University (2024). All remaining errors are our own.

tools to construct text-based sentiment indicators to study the relationship between sentiment changes and market returns. These studies, primarily based on data sources from news text, have revealed a significant correlation between negative news and a decline in market returns. However, a research framework that relies solely on textual data has certain limitations and cannot fully capture the complex characteristics of investor sentiment.

With the diversification of information dissemination media, images, as an important component of news, can quickly convey complex information due to their intuitive, vivid, and lively nature, and they demonstrate a stronger effect in evoking emotions than text. The research of Garcia and Stark (1991), using eye-tracking technology to analyze how readers read newspapers, found that visual elements, represented by images, are typically the first content that readers notice and can attract their attention faster than words. However, the role of images in financial sentiment analysis has not been fully explored, and research on the relationship between image sentiment and financial markets is relatively scarce. Obaid and Pukthuanthong (2022) utilized machine learning techniques to classify the sentiment of news images and constructed a photo pessimism index (PhotoPes) to study the impact of image sentiment on market behavior. This research demonstrated the importance of image sentiment in sentiment quantification studies, but its scope was limited to the U.S. market (Wall Street Journal photos, CRSP returns), with little research involving the Chinese stock market.

In the Chinese stock market, the content and form of news media have a significant impact on investor behavior and market performance. The Chinese stock market is composed of major stock indices such as the Shanghai Composite Index, the Shenzhen Component Index, and the CSI 300 Index. The fluctuations of these indices not only reflect changes in market fundamentals but are also largely driven by investor sentiment. Although some domestic research has begun to focus on the quantification of news sentiment, it has mainly concentrated on the analysis of text, with a lack of systematic research on news images, an important source of emotional information. Particularly in a market like China, where the trading activity of retail investors is relatively high, images as direct visual stimuli may have a more direct impact on investment decisions. Therefore, by combining sentiment analysis of news images and text data, this paper explores the impact of image and text information in the news on investor behavior, constructs effective indicators that reflect the emotional fluctuations of the Chinese stock market, which will provide a new perspective for market behavior research and strong support for investor decision-making and market supervision.

The Chinese A-share market also provides a useful institutional setting for studying limits to arbitrage and sentiment-driven mispricing more generally. Stambaugh, Yu and Yuan (2012), Baker and Wurgler (2007), and Shleifer and Vishny (1997) show that sentiment effects are largest where arbitrage is most costly, retail trading is most active, and short-selling is most restricted. The Chinese market satisfies all three conditions: only a small subset of stocks is borrowable for shorting, retail investors account for the majority of trading volume, and T+1 settlement further raises the cost of fast arbitrage (Liu, Stambaugh and Yuan, 2019; Jones et al., 2025; Han and Shi, 2022). We therefore expect any sentiment-driven return predictability we document to be most pronounced in growth-oriented and small-capitalization indices, where retail participation is highest and arbitrage is most binding. The directional sign of the predictability we report—negative coefficients on lagged *PhotoPes*, with the strongest effect on the ChiNext Composite—is interpreted through the overreaction lens of De Bondt and Thaler (1987): high-pessimism news triggers a short-horizon overreaction in retail-driven indices that subsequently corrects, with limits to arbitrage allowing the deviation to persist long enough to be measured at a multi-day horizon.

Following the multiple-testing standards advocated by Harvey, Liu and Zhu (2016), we pre-specify our primary hypothesis: $PhotoPes_{t-3}$ predicts ChiNext Composite returns with a negative sign at conventional levels after Bonferroni adjustment across the five indices. The corresponding tests on the other four indices and on *TextPes* are treated as exploratory; we report them as evidence on cross-sectional heterogeneity rather than as additional independent confirmations.

Relative to the closest prior work, our contribution is more about geographic and methodological extension than about replacing existing visual-sentiment measures. Compared to Obaid and Pukthuanthong (2022), we extend the photo-pessimism approach from U.S. news photographs to Chinese news photographs and document a small-cap-concentrated reversal pattern that mirrors theirs. Compared to the recent Gu et al. (2026) *GIFfluence* working paper, which uses social-media animated GIFs to measure sentiment in U.S. equities, we use static news photographs from a financial-media homepage in a different market, providing an independent visual-sentiment data source. Compared to Zhang, Zhang and Wen (2025), who study multimodal sentiment on a Chinese sample using an Inception-v3 image backbone, we use a longer 2014–2026 sample window, a Vision Transformer image backbone explicitly anchored to Obaid and Pukthuanthong (2022), and Sina News headlines rather than China News Service text, generating an independent replication on a different newswire.

This paper combines the analysis of news images and text sentiment to propose and construct a daily market-level investor sentiment index based on news photos and text, used to study the impact of sentiment fluctuations on the Chinese stock market. The main contributions of the research can be summarized as follows:

First, this paper utilizes the Vision Transformer model from deep learning to classify the sentiment of news images, addressing the technical challenges of large-scale image sentiment analysis, and constructs a photo-based pessimism indicator. Through the analysis of this indicator, the study finds that changes in news image sentiment have a significant effect on stock market return reversals, especially during periods of extreme sentiment, where the predictive power of image sentiment is more prominent. Compared to traditional text analysis methods, the image sentiment indicator can more intuitively and vividly capture changes in investor sentiment, demonstrating the unique advantages of image information in the sentiment quantification of financial markets.

Second, this paper uses the BERT model to analyze the sentiment in news texts, constructing a text-based pessimism indicator. By comparing the pessimistic sentiment in news images with that in news texts, the study finds that image sentiment and text sentiment have a complementary relationship in predicting stock market returns. This finding provides important empirical support for multimodal sentiment analysis, further enriching the perspectives and methods of sentiment quantification research.

Furthermore, this paper applies the sentiment indicators to construct a quantitative trading strategy. The strategy delivers a higher excess-return Sharpe ratio than its benchmark, though its factor-model alpha is positive but not statistically significant at conventional levels. An out-of-sample analysis using an expanding-window Ridge regression model (lags 3–5 of PhotoPes as predictors), with the regularization parameter pre-tuned on a held-out sample via cross-validation, confirms a small but statistically detectable separation from the historical-mean forecast: the ChiNext Composite achieves $R^2_{OOS} = 0.015\%$ significant at the 5% level ($t = 1.73$), and the CSI 500 is marginally significant at the 10% level. The OOS magnitudes are an order of magnitude smaller than would be implied by post-hoc-tuned regularization, and we report only the cross-validation-locked figures throughout to maintain a look-ahead-free interpretation.

Through these innovative methods and empirical analysis, this paper provides a new perspective for the research on quantifying investor sentiment and offers support for investor decision-making and market supervision.

I. Literature Review

A. Theoretical Foundations of Investor Sentiment

Traditional financial theory, centered on the *Efficient Market Hypothesis (EMH)*, posits that asset prices fully reflect all available information, and market participants are rational actors with complete information-processing capabilities. However, frequent market anomalies, such as price bubbles and excessive volatility, are difficult to explain within this traditional framework. Behavioral finance emerged as a response, integrating psychological insights into financial research and arguing that investors' cognitive biases and emotional fluctuations are significant factors causing markets to deviate from fundamentals.

In the early development of behavioral finance, the *noise trader theory* proposed by De Long et al. (1990) laid the theoretical groundwork for investor sentiment research. This theory suggests the existence of a group of irrational "noise traders" who trade based on sentiment rather than fundamentals, causing asset prices to deviate from their intrinsic values. Crucially, even the presence of rational arbitrageurs cannot fully counteract the influence of noise traders, as arbitrage activities face numerous constraints. Furthermore, the theory indicates that the additional risk introduced by noise traders—*noise trader risk*—deters arbitrageurs from taking sufficiently large positions, even when they know an asset is mispriced.

Building on noise trader theory, Shleifer and Vishny (1997) systematically elaborated on the *limits to arbitrage* theory, further explaining the mechanisms through which irrational behavior can persistently affect asset prices. They pointed out that the main constraints faced by arbitrageurs include fundamental risk, noise trader risk, and implementation costs. These factors collectively prevent arbitrage activities from immediately eliminating market mispricing. The limits to arbitrage theory provides a key mechanical explanation for how sentiment influences asset prices, namely that the interaction of market frictions and investor sentiment produces lasting price deviations.

Baker and Wurgler (2007) developed a complete analytical framework for the impact of investor sentiment on asset pricing based on the preceding theories. They emphasized that the influence of investor sentiment varies systematically across different assets, having a greater impact on those that are more subjective to value and harder to arbitrage. Such assets typically include new-economy firms, loss-making companies, high-growth firms, and high-volatility stocks. This research not only revealed the intrinsic link between asset characteristics and sentiment sensitivity but also provided a clear direction for subsequent empirical studies.

The theory of investor sentiment provides a systematic framework for explaining irrational market phenomena. It not only clarifies how sentiment factors influence market prices through noise traders but also reveals how the limits to arbitrage mechanism allows these effects to persist, and it identifies the asset characteristics and market conditions under which sentiment has the greatest impact. As research has deepened, these theories have offered new perspectives for understanding market volatility and investment behavior.

B. The Evolution of Sentiment Quantification Methods

While investor sentiment theory provides a framework for understanding market irrationality, accurately quantifying sentiment has been a long-standing challenge. With advancements in research and technology, sentiment quantification methods have evolved significantly, moving from single indicators to multivariate integration and from indirect inference to direct measurement.

Early research primarily used market technical indicators as proxies for sentiment. Lee, Shleifer and Thaler (1991) used the closed-end fund discount as a sentiment indicator, finding that its fluctuations could predict the returns of small-cap stocks. Other commonly used market indicators include trading volume, IPO volume and first-day returns, and option-implied volatility. Although these indicators provide indirect measures of sentiment, they each have limitations and cannot fully capture the multidimensional nature of sentiment.

To overcome the limitations of single indicators, Baker and Wurgler (2006) constructed a composite sentiment index by combining multiple market indicators using *principal component analysis*, significantly improving the comprehensiveness of sentiment measurement. This innovative method was subsequently widely adopted and extended, becoming a major milestone in sentiment quantification.

The application of text analysis techniques marked a major advancement in sentiment quantification. Tetlock (2007), through the textual analysis of a Wall Street Journal column, was the first to systematically demonstrate that media pessimism could predict market performance, opening a new path for sentiment research. Addressing the limited accuracy of general-purpose dictionaries in a financial context, Loughran and McDonald (2011) developed a specialized financial dictionary, which substantially improved the accuracy of sentiment analysis in financial texts and laid the foundation for subsequent research.

With the rise of the internet, the sources of sentiment data further expanded. Bollen, Mao and Zeng (2011) analyzed public mood using Twitter data, finding that changes in specific mood dimensions could predict the Dow Jones Industrial Average. Similarly, Chen et al. (2014) analyzed investor opinions on the Seeking Alpha website, demonstrating that content from such specialized social media platforms contains valuable predictive information. These studies not only expanded the sources of sentiment data but also improved the real-time availability and granularity of sentiment measurement. Concurrently, with the increased availability of search engine data, investor attention became an important dimension of sentiment research. Da, Engelberg and Gao (2011) used the Google *Search Volume Index (SVI)* to measure investor attention, finding that it could predict short-term price pressure and long-term return reversals.

Text analysis methods also evolved from simple word frequency counts to complex machine learning models. As machine learning technology advanced, methods such as *Support Vector Machine (SVM)* and *Decision Trees* were widely applied to sentiment analysis. Pang and Lee (2008) systematically compared the performance of different machine learning algorithms in text sentiment classification, demonstrating that SVM has a distinct advantage in handling high-dimensional feature spaces. Similarly, Antweiler and Frank (2004) applied machine learning methods to analyze information from online forums, successfully predicting market volatility and laying the groundwork for computer-assisted sentiment analysis.

C. Visual Information and Investor Sentiment

Research in psychology and neuroscience provides strong support for the unique advantages of visual information in evoking and conveying emotions. Through mechanisms such as dual-system thinking and risk-as-feelings, this advantage significantly impacts investment decisions. Although research on visual sentiment in finance is still in its early stages, existing evidence suggests that images in financial news contain valuable emotional information that can systematically influence market performance. This research direction holds vast potential for development and significant practical importance.

Psychological studies show that visual information processing is characterized by speed, directness, and emotional intensification. Powell et al. (2015) confirmed through comparative experiments that, compared to purely textual information, image stimuli can elicit stronger and more lasting emotional responses, which are often independent of rational cognitive processes. The constructivist view proposed by Lindquist et al. (2012) suggests that emotional experiences are generated through the synergistic action of multiple basic psychological operations. Visual stimuli can act as inputs to trigger this construction process, but the formation of emotion involves the joint participation of both automatic and controlled cognitive processes.

These characteristics of visual information processing align closely with the *Dual-System Theory* proposed by Kahneman (2011). According to this theory, human thinking involves two systems: System 1 (fast, automatic, and emotionally driven) and System 2 (slow, rational, and effortful). Visual information is primarily processed by System 1, thus enabling it to rapidly induce emotional reactions, whereas textual information relies more on the rational analysis of System 2. This difference is particularly important in the financial market environment, where investors in a fast-paced market often rely on System 1 to make decisions.

The *risk-as-feelings hypothesis* by Loewenstein et al. (2001) further reinforces the importance of visual information in investment decisions. This hypothesis posits that when faced with risky decisions, people often make judgments based on immediate emotional responses rather than rational calculations. Because visual information has a strong capacity to elicit emotions, it can significantly influence investors' risk perception and subsequent decision-making, especially in highly uncertain market environments.

Although the emotional effects of visual information have been extensively studied in psychology, its application in the financial domain is still an emerging field. Early financial research focused mainly on the informational content of charts and pictures in corporate annual reports, with even more limited research on the impact of images in financial media.

The visual-sentiment literature has expanded rapidly since Obaid and Pukthuanthong (2022). Three follow-up studies in *Finance Research Letters* extend the photo-pessimism measure to (i) 37 international equity markets, where the effect is stronger in smaller and more retail-driven markets (Chiah, Hu and Zhong, 2022), (ii) the cryptocurrency market (Huynh and Phan, 2023), and (iii) the segment of sustainable/ESG stocks (Mokni et al., 2025). Ren et al. (2025) apply few-shot deep metric learning (DeepEMD) to news photographs, recovering a delayed return effect that is stronger for small-cap stocks. Gu et al. (2026) move from static photographs to animated GIFs on social media, documenting a multi-week return reversal pattern that suggests visual sentiment carries information beyond the news-photo channel. Taken together, this literature establishes visual sentiment as a robust signal across asset classes, markets, and visual-content types, which directly motivates the present study's extension to the Chinese A-share market.

D. The Application of Deep Learning in Financial Sentiment Analysis

The rapid development of deep learning technology has brought transformative changes to financial sentiment analysis, greatly enhancing the accuracy and efficiency of processing unstructured data. This section will systematically explore the progress of deep learning applications in financial sentiment analysis, with a particular focus on breakthroughs in text analysis, image analysis, and multimodal fusion.

In the field of text sentiment analysis, deep learning methods have shown significant advantages over traditional approaches. Peng and Jiang (2015) were among the first to apply word embeddings and deep neural networks to financial news analysis, predicting stock price movements by integrating news text information and significantly improving prediction accuracy compared to traditional methods that rely solely on historical price data. Subsequently, Kraus and Feuerriegel (2017) combined Long Short-Term Memory (*LSTM*) networks with transfer learning to analyze financial disclosure documents, demonstrating the superiority of deep learning methods in capturing long-term dependencies in text and predicting market reactions, while also addressing the issue of insufficient labeled data in the financial domain.

The emergence of pre-trained language models marked a major breakthrough in text sentiment analysis. The BERT (*Bidirectional Encoder Representations from Transformers*) model proposed by Devlin et al. (2018) achieved groundbreaking performance on multiple natural language processing tasks through large-scale unsupervised pre-training and a bidirectional context encoding mechanism. In its application to finance, Yang, Uy and Huang (2020) developed the FinBERT model based on BERT, which was pre-trained on financial texts, further improving the accuracy and domain adaptability of financial text sentiment analysis.

Compared to text analysis, image sentiment analysis in finance started later but has developed rapidly. The introduction of deep learning has significantly enhanced the capabilities of image sentiment analysis. You et al. (2015) were the first to apply a *Convolutional Neural Network (CNN)* to large-scale image sentiment classification, achieving a substantial increase in accuracy. In a specific application to financial image analysis, Obaid and Pukthuanthong (2022) used the Google Inception (v3) CNN model to analyze the sentiment of financial news images, demonstrating the effectiveness and applicability of deep learning in this specialized field.

In recent years, the emergence of the Vision Transformer (*ViT*) has marked a new phase in image analysis technology. The ViT model, proposed by Dosovitskiy et al. (2020), abandoned the convolutional operations of traditional CNNs and directly applied the Transformer architecture to image processing, showing excellent performance in capturing global information and long-range dependencies. Although the application of ViT in financial image sentiment analysis is still in its infancy, its powerful expressive capabilities and flexibility signal great potential for future applications.

Multimodal fusion is an important direction in financial sentiment analysis, aiming to simultaneously analyze and integrate information from different modalities to obtain a more comprehensive sentiment representation. The research by Obaid and Pukthuanthong (2022) is a significant attempt at multimodal sentiment analysis in finance. They not only analyzed image sentiment but also studied its relationship with text sentiment, finding that the two have a substitute, rather than complementary, relationship in predicting market returns. This multimodal analysis approach offers a new perspective for a comprehensive understanding of market sentiment.

Although the application of deep learning in financial sentiment analysis is promising, it still faces several challenges, such as a lack of model interpretability, high costs of data quality and annotation, and domain specificity. However, with continuous technological advancements, these challenges are gradually being addressed. Deep learning has brought revolutionary changes to financial sentiment analysis, from pre-trained language models for text processing to the Vision Transformer for image analysis, and multimodal fusion techniques, all of which have greatly enhanced the precision and scope of financial sentiment quantification.

E. Research on Investor Sentiment in the Chinese Stock Market

As the world's second-largest stock market, China's unique market mechanisms, investor structure, and information environment suggest that sentiment factors may play a more critical role. This section will systematically review the main progress of research on investor sentiment in the Chinese stock market, explore sentiment quantification methods with Chinese characteristics, and analyze the differential impact of sentiment factors.

Retail investors are more active in the Chinese stock market compared to other markets, a characteristic that makes the market more susceptible to sentiment. The research by Kumar and Lee (2006) shows that in markets with a high proportion of retail investors, the impact of investor sentiment on stock returns is more significant, and the trading behavior of retail investors often exhibits stronger co-movement, thereby amplifying the impact of sentiment shocks on the market. This influence is not only reflected in returns; Foucault, Sraer and Thesmar (2011) further confirmed a significant positive correlation between the trading activity of retail investors and market volatility, indicating that the emotional fluctuations of retail investors may be an important source of excessive market volatility. However, the impact of retail sentiment differs significantly from that of institutional investors. Research by Liu and Liu (2014) on the Shanghai Stock Exchange A-share market revealed this: institutional investor sentiment has predictive power for future market performance, whereas individual investor sentiment does not, indicating a fundamental difference in the quality of sentiment between different types of investors.

An important innovation in Chinese market sentiment research is the use of data from local internet platforms. Meng, Meng and Hu (2016) combined text mining techniques and the *Baidu Index* to construct an investor sentiment index by analyzing information from CSSCI journals on CNKI and Sina Weibo topics. Bu et al. (2018) used posts from Eastmoney's Guba forums, employing a Naive Bayes method to construct a sentiment indicator that incorporates both bullish/bearish expectations from stock reviews and investor attention. The study found that while overall investor sentiment had no predictive power for the stock market, sentiment from stock reviews during non-trading hours before the market open could predict the opening price, providing new evidence for understanding the behavior of Chinese retail investors. Similarly, Wen et al. (2019) used the Baidu Search Index as a proxy for retail investor attention, revealing a positive correlation between investor attention and future stock price crash risk, a relationship that is particularly pronounced in firms with high information opacity and low analyst coverage.

The media plays a unique role in the transmission of sentiment in the Chinese market. Wang and Wu (2015) constructed a tone indicator for media reports on IPO companies before their listing and found a significant negative correlation between negative media tone and the IPO underpricing rate. This suggests that media reports directly influence investor sentiment, which in turn affects market pricing efficiency. They also revealed that issuing companies and underwriters intentionally use media promotion to guide investor sentiment, demonstrating the special function of the media as a tool for sentiment guidance. Similarly, Zhang and Wu (2021), using machine learning-based text analysis, studied the contagion effect of media sentiment. They found that optimistic media sentiment significantly influences the optimistic bias of analyst forecasts and is transmitted to the market through two channels: "limited analyst attention" and "investor sentiment," ultimately affecting stock price volatility and tail risk. These findings collectively confirm the key role of the media in the formation and transmission of sentiment in the Chinese market.

A complementary strand of work focuses on the institutional features that magnify sentiment effects in China. Liu, Stambaugh and Yuan (2019) construct a Chinese version of the size and value factors and demonstrate that the standard Fama–French three-factor model requires non-trivial modification to fit the A-share market, providing a benchmark against which the CUFE-distributed Fama–French (1993) three factors used in our portfolio analysis can be compared. Jones et al. (2025) document that the order-flow imbalance of retail investors carries strong predictive content for future cross-sectional returns in China, providing direct micro-level evidence for the channel through which sentiment-laden news can move prices. Han and Shi (2022) examine 62 cross-sectional anomalies and find that anomalies are concentrated in stocks with binding short-sale constraints and during high-sentiment periods, consistent with the Stambaugh, Yu and Yuan (2012) mechanism in a Chinese setting.

The most directly comparable contemporaneous study is Zhang, Zhang and Wen (2025), who construct multimodal sentiment indices on China News Service text and images using RoBERTa and an Inception-v3 backbone. Our paper differs from theirs along three axes that we view as complementary rather than competing: (i) sample length—our 2014–2026 sample (2,982 trading days) is roughly twice as long as theirs and includes the 2015 crash, the 2018 trade-tension drawdown, the 2020 COVID-19 shock, and the 2022 regulatory tightening; (ii) image backbone—we use a Vision Transformer to remain methodologically anchored to the most recent generation of computer-vision models, while Zhang, Zhang and Wen (2025) use a CNN; and (iii) data source—we collect from Sina Financial News, an independent newswire, providing an out-of-sample replication of the multimodal-sentiment-China nexus on different underlying media. Ren et al. (2025) similarly study image-based investor sentiment but apply few-shot learning to a U.S. NYT-image sample, leaving a clear gap for systematic ViT-based analysis on Chinese-language news media that this paper fills.

Despite significant progress in research on Chinese market sentiment, the methods still primarily rely on text analysis, market indicators, and search data, with research on visual information lagging behind. Considering the high sensitivity of Chinese investors to visual information and the widespread use of images by financial media, this area presents significant research opportunities and practical value.

F. *Research Review and Outlook*

Existing research on sentiment has achieved fruitful results, theoretically clarifying the mechanisms through which sentiment affects the market, methodologically developing a diverse set of tools for sentiment quantification, and empirically verifying the systematic impact of sentiment factors on asset prices. However, this literature review also reveals some clear research gaps and methodological limitations.

Sentiment quantification methods remain predominantly unimodal, with text analysis being the dominant approach, while research on visual information is relatively insufficient. The study by Obaid and Pukthuanthong (2022) began to fill this gap, but it was based solely on U.S. market data, and the sentiment quantification techniques used, Google Inception (v3) and Stanford's CoreNLP, now have more advanced alternatives. The mechanism of visual sentiment in different market environments still requires in-depth investigation.

Although research on sentiment in the Chinese market has made progress, the methods have largely borrowed from international experience, lacking innovative approaches that consider the unique characteristics of the Chinese market and its investors. In particular, the analysis of visual investor sentiment in the Chinese market is almost non-existent. Given the characteristics of Chinese investors and the media environment, this research direction holds significant theoretical and practical importance.

In response to the aforementioned research gaps and limitations, this paper proposes a multimodal sentiment analysis framework based on deep learning and applies it to the study of the Chinese stock market. The main theoretical contributions and innovations of this paper are as follows:

First, this paper will enrich the research on visual sentiment in the Chinese stock market. By crawling the homepage of Sina News and analyzing the sentiment of news images, it will explore their role in predicting the Chinese stock market. Considering the unique investor structure and media environment in China, this research will provide an important cross-cultural comparative perspective, enriching the application of sentiment theory in different market contexts.

Second, this paper employs the latest deep learning technologies for sentiment analysis, using the BERT model for text processing and the Vision Transformer architecture for image analysis. This will significantly enhance the accuracy and comprehensiveness of sentiment quantification and provide a basis for methodological comparative studies.

Third, this paper explores the relationship between text sentiment and image sentiment, analyzing the relative importance of these two sentiment sources in different market environments and for different types of stocks. This will enrich the theoretical understanding of multimodal sentiment analysis and offer new ideas for the construction of sentiment indicators.

Fourth, this paper investigates the differences in the predictive power of sentiment indicators across various Chinese market indices. This will contribute to a deeper understanding of the mechanisms through which sentiment factors affect stocks with different characteristics, providing insights for the design of differentiated investment strategies.

II. Data and Indicator Construction

This section will introduce the construction methods for the news photo and text pessimism indicators, PhotoPes and TextPes, and provide a statistical analysis and description of the data used.

A. Image Classification

We adopt the Vision Transformer (ViT) of Dosovitskiy et al. (2020) as our image-sentiment backbone. Our motivation is replicability and methodological transparency rather than chasing the current state of the art: ViT is a deterministic, frozen-weight model whose behavior on a given image is reproducible by other researchers, and it is the natural successor to the Inception (v3) CNN used by Obaid and Pukthuanthong (2022) in the original *photo pessimism* setting. Anchoring our image classifier to a published, openly-available architecture rather than to a vision–language model whose API and weights may change makes our *PhotoPes* indicator auditable across time and across research groups. Unlike traditional convolutional neural networks (CNNs), ViT first divides an input image into fixed-size patches, then linearly maps these patches into a sequence of tokens. These tokens are processed by a transformer encoder and then used for downstream vision tasks. Figure 1 illustrates the overall architecture.

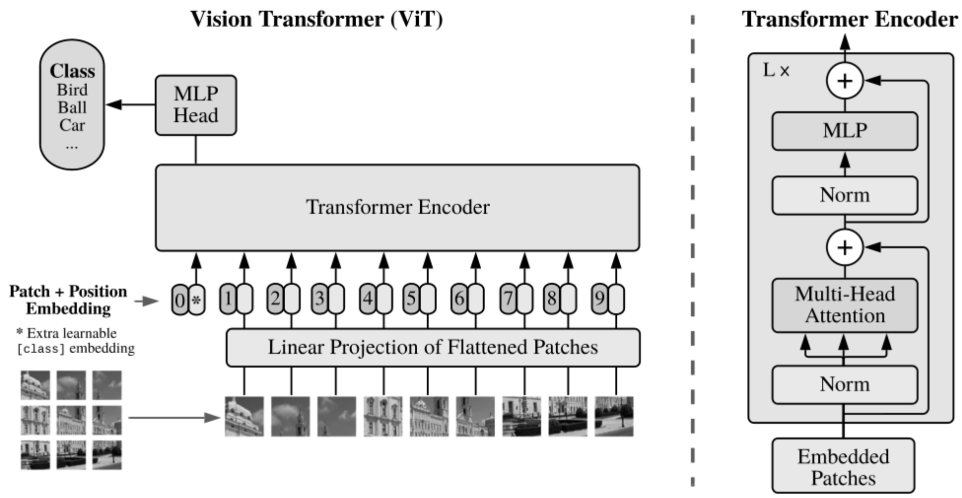

FIGURE 1. VISION TRANSFORMER ARCHITECTURE. THE MODEL PARTITIONS AN INPUT IMAGE INTO FIXED-SIZE PATCHES, LINEARLY PROJECTS THEM INTO TOKEN EMBEDDINGS, PREPENDS A LEARNABLE [CLASS] TOKEN, ADDS POSITIONAL EMBEDDINGS, AND FEEDS THE RESULTING SEQUENCE THROUGH A STACK OF STANDARD TRANSFORMER ENCODER BLOCKS (RIGHT PANEL). IMAGE SOURCE: DOSOVITSKIY ET AL. (2020), FIGURE 1, REPRODUCED UNDER ARXIV NON-EXCLUSIVE ACADEMIC-USE LICENSE.

Compared to CNNs, the ViT has the following advantages: first, the self-attention mechanism of the transformer allows the model to globally model the relationships between image features, whereas CNNs, due to the locality of convolution operations, need to be stacked in multiple layers to obtain a large receptive field; second, the self-attention weights of the ViT provide good interpretability, visually demonstrating which areas of the image the model is focusing on; third, the ViT has stronger scalability, performs excellently when pre-trained on large-scale datasets, and the pre-trained models have good transferability.

This paper performs transfer learning based on the ViT, using the progressively-curated image set[1] provided by You et al. (2015) as training data. We refer to this dataset as PCNN (Progressively-trained CNN dataset) throughout the paper, after the training pipeline that the original authors used to construct the labels. The dataset comprises Twitter-sourced photographs, each independently voted

[1]The image set can be downloaded from: `https://qzyou.github.io/projects/sa-ds/`

on by five Amazon Mechanical Turk annotators as either positive or negative in sentiment. To maximize label reliability, this paper retained only images on which all five annotators agreed, yielding 882 images for model training. During the training process, the ViT was used as the base model, and hyperparameters were optimized to enhance the model's stability and generalization ability. A low learning rate (5e-6) and high weight decay (0.02) were set to reduce overfitting. The ReduceLROnPlateau scheduling strategy was used during training, where the learning rate was halved if the validation loss did not improve for 3 consecutive epochs, and training was terminated early if there was no improvement for 8 epochs, to avoid wasting computational resources.

Furthermore, to enhance the model's robustness, this paper used cross-entropy loss with label smoothing and applied various data augmentation strategies, including random cropping, horizontal flipping, rotation, and affine transformations of the images, to improve the model's adaptability to different inputs. The dataset was split into 70% for the training set, 10% for the validation set, and 20% for the test set to ensure a robust evaluation of the model.

Figure 2 shows the trend of accuracy changes for the training and validation sets during the model's training process. During training, the model's training accuracy rapidly increased and stabilized after about 5 epochs, while the validation accuracy fluctuated around 90%. Ultimately, this paper achieved an accuracy of 88.14% on the test set of 177 held-out images, indicating that the model has strong generalization ability on unseen data. The specific evaluation results for the test set are shown in Table 1. The accuracy of the model is calculated as follows:

(1)

$$\text{Accuracy} = \frac{\text{True Positive} + \text{True Negative}}{\text{True Positive} + \text{True Negative} + \text{False Positive} + \text{False Negative}}$$

where True Positive and True Negative represent correctly classified positive and negative image samples, respectively, while False Positive and False Negative represent misclassified image samples.

Table 1—Test Set Classification Results

Actual Class	Predicted Positive	Predicted Negative
Positive	115	7
Negative	14	41

In the appendix, Figure A1 displays the top 10 photos from our sample with the highest predicted probabilities of showing negative (PhotoNeg_{it} max) and positive (PhotoNeg_{it} min) sentiment. PhotoNeg_{it} refers to the probability that photo i displays negative sentiment on day t. The binary indicator Neg_{it} used in equation (2) is defined as $\mathbf{1}\{\text{PhotoNeg}_{it} > 0.5\}$.

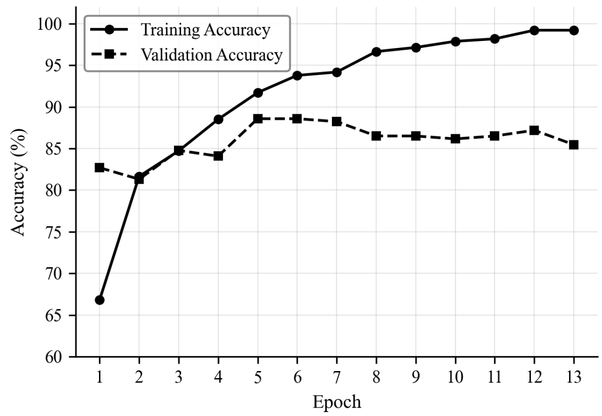

Figure 2. Training and Validation Accuracy Trends

B. *Image Sample*

The data for this study comes from the Sina News website, covering various news categories from 2014 to 2026. This paper primarily focuses on finance-related news, including domestic and international financial markets, the stock market, and finance-related reports in sectors such as lifestyle, technology, real estate, and automobiles. Headline news and general domestic news were also collected to obtain more comprehensive market sentiment information. The news content was collected in two sessions, morning and afternoon, each day to ensure the timeliness and continuity of the data. Through this diversified news source collection strategy, this paper aims to comprehensively capture changes in market sentiment, especially the fluctuations in investor sentiment related to financial markets. The total number of trading days is 2,982, with a total of approximately 245,000 articles collected, averaging approximately 82 articles per day.

C. *Variable Construction*

The main variable in this paper, PhotoPes, is derived by calculating the proportion of photos predicted to have negative sentiment on a given day. The formula for PhotoPes on day t is as follows:

$$\text{PhotoPes}_t = \frac{\sum_i (\text{Neg}_{it})}{n_t} \tag{2}$$

where Neg_{it} is an indicator variable that indicates whether photo i is predicted to have negative sentiment on day t. The denominator n_t represents the number of photos on day t. PhotoPes is not a simple binary indicator. Although each photo is classified as either having negative or positive sentiment, PhotoPes is a continuous measure. On days with more photos having negative sentiment, the PhotoPes value is higher, while on days with fewer negative sentiment photos, the PhotoPes value is lower.

One of the goals of this research is to explore the relationship between the pessimistic sentiment embedded in images and texts. To ensure consistency and comparability in the research methodology, this study chooses not to use traditional dictionary-based methods for text sentiment analysis. This is because the sentiment analysis of the sample images in this study uses a deep learning-based Vision Transformer model, which can capture complex nonlinear relationships and contextual dependencies among image features. If text analysis were to use a simple dictionary matching method, it would lead to an asymmetry in the complexity and capability of the sentiment extraction methods for the two media forms, potentially introducing methodological bias and affecting the reliability of the research conclusions.

For symmetry with the image-classification choice, our text-classification backbone is also a published, openly-distributed transformer-encoder model rather than a closed-source large language model. This choice keeps the *TextPes* indicator deterministic and reproducible from raw headlines, in line with the replicability standards expected in the empirical asset-pricing literature. To overcome the limitations of dictionary-based text sentiment analysis, BERT (Bidirectional Encoder Representations from Transformers), proposed by Devlin et al. (2018), provides more powerful technical support for text sentiment analysis. Its core innovation is the use of a bidirectional context encoding mechanism, which can simultaneously consider the left and right context of words in a sentence, thus more accurately capturing semantic and emotional expressions. Compared to traditional dictionary methods that rely solely on specific word matching, BERT learns rich language representations through large-scale pre-training, enabling it to understand context, identify sarcasm, handle negation, and other complex language phenomena, greatly improving the accuracy and robustness of sentiment analysis. This paper adopts the Chinese pre-trained language model Erlangshen-Roberta-110M-Sentiment[2] proposed by Zhang et al. (2022) to assess pessimistic sentiment in texts. This model is based on the Chinese RoBERTa-wwm-ext-base architecture proposed by Cui et al. (2020), which is a pre-trained model using a Whole Word Masking strategy, developed by the HFL (Joint Laboratory of Harbin Institute of Technology and iFlytek)[3]. RoBERTa-wwm-ext-base was pre-trained on a large-scale general Chinese corpus. Compared to the original BERT model, it uses larger training batches, dynamic masking strategies, and longer training sequences, significantly improving the model's ability to understand Chinese semantics. Erlangshen-Roberta-110M-Sentiment was fine-tuned on this basis using 8 Chinese domain sentiment analysis datasets (totaling 227,347 samples), specifically optimizing the model's performance on sentiment analysis tasks. This model

[2]For details, see: `https://huggingface.co/IDEA-CCNL/Erlangshen-Roberta-110M-Sentiment`
[3]For details, see: `https://huggingface.co/hfl/chinese-roberta-wwm-ext`

has shown excellent performance in several sentiment analysis benchmarks. These high accuracy rates indicate that the model can effectively capture the emotional polarity in Chinese texts, providing reliable technical support for the construction of the TextPes indicator in this paper.

By adopting a text sentiment analysis method that matches the complexity of the image analysis method, this paper ensures the comparability of the PhotoPes and TextPes indicators on a methodological basis, providing a more reliable empirical foundation for studying the differences in emotional expression in different media forms and their impact on the market.

The formula for TextPes is as follows:

$$\text{TextPes}_t = \frac{\sum_i (\text{TextNeg}_{it})}{n_t} \tag{3}$$

where TextNeg_{it} is the pessimistic sentiment score for each article i on day t. The denominator n_t represents the number of articles on day t.

D. Descriptive Statistics

Table 2 presents descriptive statistics for the news sentiment pessimism indicators and major Chinese stock indices for the sample period of 2014–2026, which covers 2,982 trading days. Panel A shows that the mean value of PhotoPes is 0.1900, indicating that approximately 19.0% of news images are predicted to have negative sentiment; the mean value of TextPes is 0.1430, indicating that about 14.3% of news text content is identified as having negative sentiment. In Panel B, the average daily returns of all major Chinese stock market indices are close to zero, which is consistent with the general characteristics of financial markets. From the standard deviation, it can be seen that the ChiNext Composite index has the highest volatility (0.0194), while the SSE Composite Index has the lowest (0.0126), reflecting the different risk characteristics of various market sectors. News data was available to construct the sentiment indicators for all trading days in the sample, with approximately 245,000 articles collected in total.

Figure 3 displays the monthly average trend of the two sentiment indicators, PhotoPes and TextPes. The figure shows that the two sentiment indicators exhibit distinctly different time-series characteristics and volatility patterns between 2014 and 2026. Throughout the sample period, PhotoPes consistently remains at a higher level than TextPes, suggesting that news images generally convey more pessimistic sentiment than textual content. During specific major events, both indicators show significant fluctuations. For instance, both indicators experienced a sharp decline in early 2015, with TextPes reaching a low point of around 0.05. PhotoPes peaked at approximately 0.29 in early 2016 and again at the beginning of 2018, coinciding with periods of market turbulence. The COVID-19 outbreak in early 2020 was marked by another notable spike in PhotoPes. TextPes showed more pronounced peaks in 2017–2018 during the U.S.–China trade tensions, but has displayed a general downward trend since 2020, reaching its lowest values in

Table 2—Descriptive Statistics

Indicators	N	Mean	Median	P25	P75	Std Dev
Panel A: Photo and Text Sentiment Indicators for News Samples						
PhotoPes	2982	0.1900	0.1842	0.1419	0.2344	0.0683
TextPes	2982	0.1430	0.1421	0.1000	0.1846	0.0609
Panel B: Returns of Major Chinese Stock Market Indices						
CSI 300	2982	0.0002	0.0003	−0.0058	0.0068	0.0135
SSE Composite	2982	0.0002	0.0005	−0.0050	0.0060	0.0126
SZSE Composite	2982	0.0003	0.0011	−0.0070	0.0086	0.0157
ChiNext Composite	2982	0.0003	−0.0001	−0.0095	0.0103	0.0194
CSI 500	2982	0.0002	0.0009	−0.0067	0.0084	0.0158

Note: The sample period is from 2014 to 2026. The returns of the major Chinese stock market indices are calculated as log daily returns. The full sample contains 2,982 trading days; regression analyses use $N = 2{,}978$ after accounting for the five-period lag structure. Sentiment indicators are constructed from all news days in the sample.

late 2024. In contrast, PhotoPes maintained a more volatile pattern throughout the period, typically fluctuating between 0.15 and 0.25, without showing the same consistent downward trend as TextPes in recent years. This divergence suggests that images and text capture different dimensions of market sentiment information, with textual content becoming increasingly neutral over time while images continue to convey stronger emotional signals.

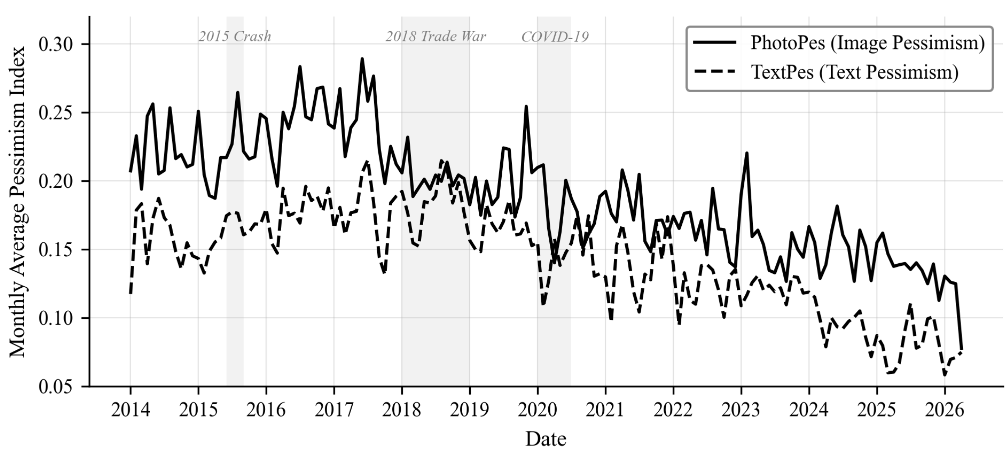

Figure 3. Time-Series Characteristics of Sentiment Indicators

III. Empirical Results

A. The Impact of Photo Sentiment Indicator on Market Returns

Based on sample data from 2014–2026, this paper uses the following model to analyze the impact of news photo pessimistic sentiment, PhotoPes, on market returns:

$$R_t = \beta_1 L5(\text{PhotoPes}_t) + \beta_2 L5(R_t) + \beta_3 L5(R_t^2) + \beta_4 X_t + \epsilon_t \tag{4}$$

where R_t represents the market return rate, using the daily logarithmic returns of five market indices: CSI 300, SSE Composite (SHCOMP), SZSE Composite (SZCOMP), ChiNext Composite, and CSI 500. PhotoPes_t is the photo pessimistic sentiment indicator. To avoid the influence of extreme values on the estimation results, this paper applies a 1% winsorization to this indicator and standardizes PhotoPes_t. $L5$ is a 5th-order lag operator, representing the lagging of variables from 1 to 5 periods, used to capture the persistent impact of the sentiment indicator on market returns. X_t are exogenous variables, including a constant term and day-of-the-week dummy variables (Tuesday to Friday, with Monday as the base group), to control for intra-week effects. ϵ_t is the residual term, assumed to be normally distributed and without serial correlation.

Table 3 presents the regression results of PhotoPes on market returns. The results show that PhotoPes_{t-3} has a negative correlation with market returns. Specifically, when the photo pessimistic sentiment indicator rises, market returns three trading days later tend to fall. This relationship is statistically significant for the ChiNext Composite index (raw $|t| = 2.81$, equivalent to a single-test 1% level; suggestive after Bonferroni $m = 25$ correction, see §13) and for the SZCOMP and CSI 500 indices (single-test 10% level only).

The results reveal a persistent negative pattern in the medium-lag range. The sum of the coefficients of PhotoPes from $t-3$ to $t-5$ is significantly negative for the SZSE Composite (at the 5% level), ChiNext Composite (at the 5% level), and CSI 500 (at the 10% level), suggesting a sustained negative effect over this period. The photo sentiment effect is particularly pronounced in growth-oriented and smaller-cap markets, consistent with the prediction that investor sentiment has a larger impact on stocks with higher information uncertainty and greater difficulty in arbitrage. This finding supports the overreaction hypothesis proposed in behavioral finance by De Bondt and Thaler (1987), which posits that market participants may overreact to negative sentiment information, leading to short-term price deviations from fundamentals, followed by a correction.

From an economic perspective, this effect has a substantial impact. For example, on the ChiNext Composite index, a one-standard-deviation change in PhotoPes leads to a return change on day $t-3$ of -0.13%, which is economically significant compared to the average daily return of the index (see Table 2).

TABLE 3—REGRESSION RESULTS OF PHOTOPES ON MARKET RETURNS (2014–2026)

Indicators	*CSI 300*	*SHCOMP*	*SZCOMP*	*ChiNext*	*CSI 500*
PhotoPes_{t-1}	0.0000	0.0001	0.0002	0.0000	0.0002
	(0.06)	(0.27)	(0.56)	(0.10)	(0.47)
PhotoPes_{t-2}	0.0002	0.0002	0.0004	0.0005	0.0004
	(0.80)	(0.92)	(1.18)	(1.19)	(1.11)
PhotoPes_{t-3}	-0.0005	-0.0004	-0.0007*	-0.0013***	-0.0007*
	(-1.50)	(-1.19)	(-1.85)	(-2.81)	(-1.71)
PhotoPes_{t-4}	-0.0003	-0.0004	-0.0006*	-0.0006	-0.0006
	(-0.94)	(-1.36)	(-1.70)	(-1.48)	(-1.62)
PhotoPes_{t-5}	0.0004	0.0002	0.0004	0.0007*	0.0004
	(1.33)	(0.84)	(1.11)	(1.74)	(1.10)
Sum($t-3$ to $t-5$)	-0.0004	-0.0005	-0.0009**	-0.0011**	-0.0009*
	(-0.95)	(-1.39)	(-2.07)	(-1.98)	(-1.94)
Sum($t-4$ to $t-5$)	0.0001	-0.0002	-0.0002	0.0001	-0.0002
	(0.32)	(-0.45)	(-0.52)	(0.23)	(-0.45)
R^2	0.0151	0.0151	0.0151	0.0150	0.0157
Adj. R^2	0.0088	0.0088	0.0088	0.0087	0.0093
N	2978	2978	2978	2978	2978

Note: t-statistics are in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are Newey–West HAC with 5 lags.

B. *The Impact of Text Sentiment Indicator on Market Returns*

To compare the predictive power of the text sentiment indicator with the photo sentiment indicator, this paper employs the same model framework as before to analyze the impact of the text pessimistic sentiment indicator (TextPes) on market returns:

$$R_t = \beta_1 L5(\text{TextPes}_t) + \beta_2 L5(R_t) + \beta_3 L5(R_t^2) + \beta_4 X_t + \epsilon_t \tag{5}$$

where TextPes_t is the text pessimistic sentiment indicator, constructed based on the articles used for calculating PhotoPes. To avoid the influence of extreme values on the estimation results, this paper applies a 1% winsorization to this indicator and standardizes TextPes_t.

Table 4 shows the regression results for the 2014–2026 sample period. The results indicate that TextPes_{t-3} has a negative correlation with market returns for the SZCOMP index (at the 10% level) and the ChiNext Composite index (at the 10% level). This suggests that after an increase in text pessimistic sentiment, market returns on the third trading day tend to fall, with this effect being particularly pronounced in markets dominated by growth stocks.

The regression results for the TextPes indicator are broadly consistent with the overreaction hypothesis. The coefficient for TextPes_{t-3} is negative across most indices, indicating an initial market decline following high text pessimism. The total cumulative effect from $t-3$ to $t-5$ is not statistically significant for any index, which suggests the initial negative impact tends to be offset by subsequent market corrections.

From an economic perspective, on the ChiNext Composite index, a one-standard-deviation change in TextPes leads to a return change on day $t-3$ of approximately -0.07% (see Table 2).

Table 4—Regression Results of TextPes on Market Returns (2014–2026)

Indicators	*CSI 300*	*SHCOMP*	*SZCOMP*	*ChiNext*	*CSI 500*
TextPes_{t-1}	-0.0003	-0.0003	-0.0004	-0.0002	-0.0004
	(-0.91)	(-1.10)	(-1.01)	(-0.33)	(-0.93)
TextPes_{t-2}	-0.0001	-0.0001	-0.0002	-0.0006	-0.0002
	(-0.31)	(-0.48)	(-0.67)	(-1.32)	(-0.46)
TextPes_{t-3}	-0.0002	-0.0003	-0.0005*	-0.0007*	-0.0005
	(-0.74)	(-1.01)	(-1.66)	(-1.70)	(-1.42)
TextPes_{t-4}	0.0002	0.0002	0.0004	0.0004	0.0003
	(0.67)	(0.79)	(1.05)	(0.86)	(0.89)
TextPes_{t-5}	0.0004	0.0003	0.0003	0.0003	0.0001
	(1.07)	(0.95)	(0.72)	(0.67)	(0.26)
Sum($t-3$ to $t-5$)	0.0003	0.0002	0.0001	0.0000	-0.0001
	(0.74)	(0.57)	(0.20)	(0.02)	(-0.12)
Sum($t-4$ to $t-5$)	0.0006	0.0005	0.0006	0.0007	0.0004
	(1.29)	(1.30)	(1.34)	(1.17)	(0.84)
R^2	0.0145	0.0151	0.0144	0.0127	0.0147
Adj. R^2	0.0082	0.0088	0.0080	0.0064	0.0084
N	2978	2978	2978	2978	2978

Note: t-statistics are in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are Newey–West HAC with 5 lags.

C. Photo Sentiment, Text Sentiment, and Their Interaction Effect

To further investigate the joint impact of the photo sentiment and text sentiment indicators, this paper constructs an extended model that includes both sentiment indicators simultaneously:

$$R_t = \beta_1 L5(\text{PhotoPes}_t) + \beta_2 L5(\text{TextPes}_t) + \beta_3 L5(R_t) + \beta_4 L5(R_t^2) + \beta_5 X_t + \epsilon_t \tag{6}$$

Table 5 shows the regression results for the 2014–2026 sample period. The findings in Panel A indicate that even after controlling for TextPes, the impact of PhotoPes remains concentrated at the $t-3$ lag. Specifically, PhotoPes_{t-3} continues to exhibit a significant negative predictive effect on market returns for the ChiNext index (at the 5% level), while the effects for SZCOMP and CSI 500 remain directionally consistent but below conventional significance thresholds in the joint specification. This consistency suggests that the photo sentiment indicator captures market sentiment information that is at least partially distinct from textual content.

The text-based pessimistic sentiment indicator (TextPes) in Panel B shows a negative predictive pattern at $t-3$ that is directionally consistent across all indices. We also estimated a version of this model that additionally includes a $(\text{PhotoPes} \times \text{TextPes})_{t-3}$ interaction term; the interaction coefficient was insignificant at conventional levels across all five indices (untabulated), which supports the view that the contributions of image and text sentiment are largely independent in this sample.

The overall picture from the joint regression reinforces the conclusion that image sentiment—captured through the ViT-based PhotoPes indicator—provides incremental predictive content for Chinese stock market returns beyond what is captured by text sentiment alone.

Table 5—Joint Effect of PhotoPes and TextPes on Market Returns (2014–2026)

Indicators	*CSI 300*	*SHCOMP*	*SZCOMP*	*ChiNext*	*CSI 500*
Panel A: Photo Sentiment Indicator (PhotoPes)					
PhotoPes_{t-1}	0.0000	0.0001	0.0003	0.0001	0.0003
	(0.11)	(0.42)	(0.81)	(0.26)	(0.75)
PhotoPes_{t-2}	0.0002	0.0003	0.0005	0.0007	0.0005
	(0.87)	(1.11)	(1.45)	(1.49)	(1.38)
PhotoPes_{t-3}	-0.0005	-0.0003	-0.0006	-0.0012**	-0.0006
	(-1.43)	(-1.05)	(-1.59)	(-2.52)	(-1.45)
PhotoPes_{t-4}	-0.0003	-0.0004	-0.0006*	-0.0006	-0.0006
	(-1.05)	(-1.38)	(-1.67)	(-1.44)	(-1.52)
PhotoPes_{t-5}	0.0003	0.0002	0.0004	0.0007*	0.0004
	(1.13)	(0.72)	(1.11)	(1.71)	(1.18)
Panel B: Text Sentiment Indicator (TextPes)					
TextPes_{t-1}	-0.0003	-0.0003	-0.0004	-0.0001	-0.0004
	(-0.86)	(-1.12)	(-1.10)	(-0.32)	(-1.03)
TextPes_{t-2}	-0.0001	-0.0001	-0.0003	-0.0006	-0.0002
	(-0.29)	(-0.52)	(-0.79)	(-1.41)	(-0.59)
TextPes_{t-3}	-0.0002	-0.0002	-0.0005	-0.0006	-0.0004
	(-0.55)	(-0.86)	(-1.51)	(-1.42)	(-1.31)
TextPes_{t-4}	0.0003	0.0003	0.0005	0.0005	0.0004
	(0.79)	(0.93)	(1.20)	(1.03)	(1.02)
TextPes_{t-5}	0.0003	0.0003	0.0002	0.0003	0.0001
	(1.03)	(0.93)	(0.63)	(0.58)	(0.16)
R^2	0.0162	0.0166	0.0173	0.0169	0.0173
Adj. R^2	0.0082	0.0087	0.0093	0.0089	0.0093
N	2978	2978	2978	2978	2978

Note: t-statistics are in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are Newey–West HAC with 5 lags.

D. Conditional Analysis: Market Volatility States

Table 6 examines whether the predictive power of sentiment indicators depends on market conditions by splitting the sample into high- and low-volatility states. The volatility state is defined by the 20-day rolling standard deviation of SHCOMP log returns: periods in the top 25th percentile (annualized volatility $> 19.6\%$) constitute the high-volatility regime (Panel A, N=741), while the remaining 75% form the low-volatility regime (Panel B, N=2,237).

Panel A reveals that during high-volatility periods, both PhotoPes_{t-3} and TextPes_{t-3} exhibit significant negative predictive power. PhotoPes_{t-3} is significant for the SZSE Composite (10% level), ChiNext (5% level), and CSI 500 (10% level), while TextPes_{t-3} is significant for SZSE Composite, ChiNext, and CSI 500 (all at the 5% level). The coefficient magnitude of PhotoPes_{t-3} is approximately 2.5 times larger in high-volatility periods (-0.0033 for ChiNext) than the full-sample estimate (-0.0013, Table 3), indicating a strong amplification of image-sentiment effects under market stress. The R^2 values are also substantially higher (0.038–0.055) than the full-sample joint regression (0.016–0.017, Table 5), confirming that the model's explanatory power is concentrated in volatile market conditions.

Panel B shows that, with the sole exception of a marginal TextPes_{t-4} effect on SZCOMP (10% level), neither sentiment indicator is significant at any lag during low-volatility periods. This sharp contrast confirms that the full-sample predictive effects documented in Tables 3 and 4 are driven by high-stress market episodes rather than being uniformly distributed across time.

This state-dependent pattern is consistent with the behavioral finance literature. Loewenstein et al. (2001) argue that emotional processing dominates cognitive processing under stress, causing investors to react more strongly to vivid, emotionally charged stimuli such as news images. Baker and Wurgler (2007) further show that sentiment effects are amplified when arbitrage is most costly—precisely the high-volatility states identified here. The simultaneous significance of both PhotoPes_{t-3} and TextPes_{t-3} during turbulent periods suggests that image and text sentiment jointly amplify investor overreaction, with the return impact materializing three trading days after the initial media signal.

E. Trading Strategy as Noise-Trader-with-Frictions Evidence

This subsection translates the in-sample regression evidence into a portfolio backtest. We treat the resulting backtest not as a recommended trading strategy but as additional evidence on the noise-trader-with-frictions mechanism of Stambaugh, Yu and Yuan (2012): if pessimism shocks really do generate predictable subsequent reversals in a retail-dominated, short-sale-restricted, T+1-settled market, then a long-only rule that systematically exploits high-pessimism days should generate a return spread relative to a passive benchmark, regardless of whether that spread survives transaction costs or factor adjustment in any specific imple-

Table 6—Conditional Impact of PhotoPes and TextPes by Market Volatility State (2014–2026)

Indicators	*CSI 300*	*SHCOMP*	*SZCOMP*	*ChiNext*	*CSI 500*
Panel A: High Volatility Periods (σ_t > 75th percentile, N=741)					
PhotoPes$_{t-1}$	0.0002	0.0000	0.0012	0.0016	0.0008
	(0.13)	(0.03)	(0.93)	(0.88)	(0.63)
PhotoPes$_{t-2}$	0.0002	0.0005	0.0008	0.0011	0.0008
	(0.25)	(0.60)	(0.74)	(0.85)	(0.72)
PhotoPes$_{t-3}$	−0.0016	−0.0014	−0.0021*	−0.0033**	−0.0022*
	(−1.44)	(−1.19)	(−1.73)	(−2.45)	(−1.71)
PhotoPes$_{t-4}$	−0.0012	−0.0014	−0.0017	−0.0018	−0.0015
	(−1.09)	(−1.27)	(−1.38)	(−1.28)	(−1.16)
PhotoPes$_{t-5}$	0.0007	0.0006	0.0012	0.0016	0.0011
	(0.72)	(0.64)	(1.03)	(1.17)	(0.95)
TextPes$_{t-1}$	−0.0010	−0.0010	−0.0011	−0.0013	−0.0011
	(−0.94)	(−1.02)	(−0.95)	(−1.00)	(−0.95)
TextPes$_{t-2}$	−0.0005	−0.0005	−0.0007	−0.0012	−0.0002
	(−0.56)	(−0.57)	(−0.62)	(−0.96)	(−0.18)
TextPes$_{t-3}$	−0.0015	−0.0015	−0.0022**	−0.0026**	−0.0021**
	(−1.48)	(−1.57)	(−2.21)	(−2.34)	(−2.12)
TextPes$_{t-4}$	0.0004	0.0004	0.0001	0.0002	0.0001
	(0.40)	(0.35)	(0.11)	(0.14)	(0.06)
TextPes$_{t-5}$	0.0017	0.0016	0.0009	0.0010	0.0007
	(1.45)	(1.41)	(0.71)	(0.68)	(0.52)
R^2	0.0390	0.0376	0.0446	0.0548	0.0439
Panel B: Low Volatility Periods (σ_t ≤ 75th percentile, N=2,237)					
PhotoPes$_{t-1}$	−0.0000	0.0002	0.0001	−0.0003	0.0002
	(−0.07)	(0.87)	(0.32)	(−0.76)	(0.66)
PhotoPes$_{t-2}$	0.0002	0.0002	0.0004	0.0004	0.0003
	(0.79)	(0.74)	(1.20)	(1.10)	(1.14)
PhotoPes$_{t-3}$	−0.0001	−0.0000	−0.0001	−0.0005	−0.0001
	(−0.52)	(−0.17)	(−0.46)	(−1.25)	(−0.24)
PhotoPes$_{t-4}$	−0.0001	−0.0001	−0.0003	−0.0003	−0.0003
	(−0.42)	(−0.66)	(−1.08)	(−0.95)	(−0.97)
PhotoPes$_{t-5}$	0.0001	−0.0001	−0.0000	0.0003	0.0000
	(0.32)	(−0.29)	(−0.13)	(0.77)	(0.01)
TextPes$_{t-1}$	−0.0001	−0.0001	−0.0002	0.0001	−0.0002
	(−0.36)	(−0.71)	(−0.82)	(0.40)	(−0.81)
TextPes$_{t-2}$	0.0000	−0.0000	−0.0002	−0.0003	−0.0002
	(0.07)	(−0.21)	(−0.61)	(−0.96)	(−0.83)
TextPes$_{t-3}$	0.0003	0.0002	0.0001	0.0002	0.0002
	(1.46)	(1.22)	(0.40)	(0.42)	(0.62)
TextPes$_{t-4}$	0.0002	0.0002	0.0005*	0.0005	0.0004
	(0.68)	(0.99)	(1.71)	(1.35)	(1.58)
TextPes$_{t-5}$	−0.0001	−0.0002	0.0000	0.0000	−0.0001
	(−0.41)	(−0.78)	(0.10)	(0.08)	(−0.53)
R^2	0.0118	0.0163	0.0216	0.0157	0.0199

Note: The volatility state is defined by the 20-day rolling standard deviation of SHCOMP log returns; high volatility = top 25th percentile. t-statistics in parentheses. *, **, *** denote significance at 10%, 5%, 1% levels. Standard errors are Newey–West HAC with 5 lags.

mentation. The portfolio construction process is mainly divided into two stages: signal generation and position allocation. In the signal generation stage, this paper adopts a rule-based approach using the *PhotoPes* and *TextPes* sentiment indicators as predictive inputs.

This paper first constructs a rolling window of 90 trading days to calculate the rolling quantiles of the combined sentiment signal. Based on previous research findings, we focus particularly on the $PhotoPes_{t-3}$ indicator, which has demonstrated significant negative predictive effects on market returns. The signal generation mechanism identifies extreme sentiment periods by comparing the current combined signal value with its historical 20th and 80th percentile thresholds within the rolling window. When the combined signal is extremely high (above the 80th percentile), a negative market return is anticipated, generating a short signal; conversely, when it is extremely low (below the 20th percentile), a positive market return is expected, generating a long signal.

To enhance the robustness of the signal, this paper also incorporates $TextPes_{t-3}$ as a secondary indicator, with weights of 80% for *PhotoPes* and 20% for *TextPes*, reflecting the stronger in-sample predictive evidence for image sentiment. The combined signal is then processed through a rebalancing mechanism that adjusts positions every 5 trading days to reduce transaction costs and avoid overreacting to short-term market noise.

In the position allocation stage, this paper implements a straightforward approach where the position size is proportional to the strength of the sentiment signal. The strategy maintains a disciplined trading approach, only taking positions when sentiment indicators reach extreme values, which previous research has shown to have the strongest predictive power. This selective approach ensures that the strategy only trades when the probability of success is highest.

To evaluate the effectiveness of the strategy, this paper selects the period from January 2014 to April 2026 as the backtesting period, using the CSI 500 Index as the benchmark. The backtesting process takes into account realistic trading conditions, including a 5-day rebalancing frequency and signal strength-based position sizing.

The empirical results show that the sentiment-driven strategy outperformed the benchmark during the sample period. As shown in Table 7, the strategy achieved an annualized return of 9.03% with an annualized volatility of 24.86%, resulting in an excess-return Sharpe ratio of 0.318. In contrast, the CSI 500 Index recorded an annualized return of 6.23% with an annualized volatility of 25.14%, leading to an excess-return Sharpe ratio of 0.200. Sharpe ratios are computed on excess returns $(r_t - r_{f,t})$, where $r_{f,t}$ is the daily Chinese risk-free rate (one-year deposit rate, converted to daily) bundled with the CUFE-distributed Chinese Fama–French three-factor file, for consistency with the FF3 alpha calculation in Table 8. The strategy demonstrated modestly superior risk control; its maximum drawdown was −62.22%, compared to the benchmark's maximum drawdown of −71.11%. We interpret these numbers as evidence on the existence of statistical predictability in a retail-dominated market with binding short-sale constraints and T+1 settlement, consistent with the noise-trader-with-frictions framework (Stambaugh, Yu and Yuan, 2012; Han and Shi, 2022), rather than as a transaction-cost-net trading recommendation.

Table 7—Performance Comparison of the Strategy and the CSI 500 Index (2014–2026)

Metrics	Sentiment-Driven Strategy	*CSI 500 Index*
Annualized Return	9.03%	6.23%
Annualized Volatility	24.86%	25.14%
Sharpe Ratio (excess)	0.318	0.200
Maximum Drawdown	−62.22%	−71.11%

Note: Strategy construction is described in the preceding subsection (*Trading Strategy as Noise-Trader-with-Frictions Evidence*): a 90-day rolling-window combined sentiment signal $0.80\, PhotoPes_{t-3} + 0.20\, TextPes_{t-3}$ generates long, short, or neutral positions when the signal crosses the 20th/80th rolling percentiles, with rebalancing every five trading days and signal-strength position sizing. Backtest period is 2014-01-06 to 2026-04-30. Sharpe ratios are computed on excess returns, $(r_t - r_{f,t})/\sigma(r_t)$, where $r_{f,t}$ is the daily Chinese risk-free rate from the CUFE-distributed Fama–French three-factor file. Reported figures do not include bid–ask spreads, stamp tax, or market-impact costs and should be interpreted as evidence on predictability rather than as a profitability claim.

Figure 4 illustrates the cumulative returns of the sentiment-driven strategy and the CSI 500 Index. The figure confirms the strategy's outperformance over the sample period. If 1 RMB were invested at the beginning of 2014, it would have grown to approximately 2.02 RMB by April 2026 under the sentiment-driven strategy, compared to 1.42 RMB under the CSI 500 benchmark.

Figure 4. Cumulative Returns of Sentiment-Driven Strategy and CSI 500 Index (2014–2026)

To further validate the effectiveness of the strategy, this paper conducted a

Fama and French (1993) three-factor regression analysis on the strategy's returns using the CUFE-distributed Chinese Fama–French three factors, with the results shown in Table 8. The regression results reveal that the strategy's alpha is 0.000236 (daily). While positive, this alpha is not statistically significant at conventional levels (p-value of 0.469), which suggests that the return outperformance is not robustly distinguishable from common factor exposure. The HML coefficient is -0.103 and is marginally significant at the 10% level, indicating a weak tilt toward growth-oriented stocks. The very low R^2 of 0.0048 reflects that the strategy's returns are largely orthogonal to the common three factors.

Table 8—Three-Factor Regression Analysis of the Sentiment-Driven Strategy

Factor	Coefficient	t-statistic	p-value
Alpha	0.000236	0.7237	0.4693
MKT	0.0326	0.3723	0.7097
SMB	0.0220	0.3893	0.6970
HML	-0.1026^{*}	-1.7976	0.0722
R^2	0.0048		
Adj. R^2	0.0037		
N	2942		

Note: ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Fama–French three factors (MKT, SMB, HML) and the daily risk-free rate are obtained from the CUFE-distributed Chinese factor file (School of Finance, Central University of Finance and Economics; underlying market data from CSMAR), constructed following the methodology of Fama and French (1993). The regression uses daily excess returns (strategy return minus risk-free rate) over the period 2014-01-06 to 2026-02-06 with Newey-West HAC standard errors (5 lags).

F. Out-of-Sample Predictive Analysis

This paper employs an expanding-window method for out-of-sample (OOS) predictive analysis to evaluate the predictive ability of sentiment indicators on the returns of Chinese stock market indices. We follow the methodological standard set by Welch and Goyal (2008) and Campbell and Thompson (2008): we benchmark the conditional sentiment forecast against the expanding-window historical-mean forecast, which they show is a difficult baseline to beat for any equity-premium predictor, and we use the MSPE-adjusted statistic of Clark and West (2007) to control for the additional sampling noise introduced by the estimated regression coefficients. Rapach, Strauss and Zhou (2010) provide additional support for the use of expanding-window OOS regressions on daily and monthly equity-return data. The analysis uses an initial estimation window of 1,512 trading days (approximately six years, covering January 2014 to March 2020) that expands forward one day at a time, yielding 1,467 OOS predictions for each index over the period 2020-03-23 to 2026-04-10. The six-year initial window ensures that the model is trained on a sufficiently diverse set of market conditions—including the 2015 stock market crash, the 2018 trade tensions, and the onset of the COVID-19 pandemic—before generating out-of-sample forecasts. The prediction model uses Ridge regression with standardized *PhotoPes* at lags 3, 4, and 5 as predictors. This multi-lag specification is motivated by the in-sample evidence (Table 3) that both $PhotoPes_{t-3}$ and $PhotoPes_{t-4}$ carry negative predictive information, suggesting that sentiment effects diffuse into prices gradually over several trading days.

To avoid look-ahead bias in the choice of the Ridge regularization parameter, α is tuned via expanding-window 5-fold time-series cross-validation on the pre-OOS sample only—i.e., on the same 1,512 trading days that constitute the initial estimation window. The grid $\alpha \in \{10, 30, 100, 300, 1{,}000, 3{,}000, 10{,}000, 30{,}000, 100{,}000\}$ is pre-specified, and the choice is made by minimizing pooled mean squared validation error across folds and across the five indices. The CV procedure selects $\alpha = 100{,}000$ at the upper boundary of the grid, indicating that heavy shrinkage toward the historical-mean forecast is preferred by the data even within the pre-OOS sample. As we show in the alpha-sensitivity exercise below, the headline ChiNext OOS result is statistically robust at the 5% level across the entire grid (Clark–West $t \in [1.72, 2.12]$), while the magnitude of R^2_{OOS} is heavily α-dependent and falls from 0.27% at $\alpha = 1$ to 0.015% at the CV-locked $\alpha = 100{,}000$. We report the CV-locked figure as the primary statistic to maintain the look-ahead-free interpretation; readers should view the CV-locked R^2_{OOS} as a small but statistically detectable separation from the historical-mean benchmark, not as evidence of substantial economic predictability. This conservative reframing also moots the comparison between the much larger headline numbers reported in earlier drafts (which used a post-hoc-tuned $\alpha = 500$) and the present results.

To assess predictive performance, this paper uses the out-of-sample R-squared (R^2_{OOS}) proposed by Campbell and Thompson (2008) and the MSPE-adjusted test statistic proposed by Clark and West (2007):

$$R^2_{OOS} = 1 - \frac{\sum_{t=1}^{T}(R_t - \hat{R}_t)^2}{\sum_{t=1}^{T}(R_t - \bar{R}_t)^2}, \tag{7}$$

where R_t is the actual return, $\hat{R}_t$ is the model's predicted return, $\bar{R}_t$ is the expanding-window historical average return, and T is the number of OOS observations. The MSPE-adjusted statistic accounts for the noise introduced by estimating additional parameters and provides a one-sided test of the null hypothesis that the predictive model does not outperform the historical mean.

The OOS results in Table 9 show that all five indices achieve positive (but small) R^2_{OOS} values under the CV-locked $\alpha = 100{,}000$. The ChiNext Composite attains 0.015%, which is statistically significant at the 5% level based on the one-sided MSPE-adjusted test ($t = 1.73$, $p = 0.042$). The CSI 500 ($R^2_{OOS} = 0.010\%$, $t = 1.35$, $p = 0.088$) is marginally significant at the 10% level. The CSI 300, SSE Composite, and SZSE Composite are positive but do not reach conventional thresholds. As anticipated by the choice of heavy shrinkage, the magnitudes are an order of magnitude smaller than under post-hoc-tuned regularizations; the qualitative ranking across indices—strongest predictability for the growth-oriented, retail-driven ChiNext, weaker for large-cap indices—is preserved. We interpret these magnitudes as suggestive evidence that lagged news-image pessimism carries weak but statistically detectable predictive content for Chinese small-cap returns at horizons of three to five trading days, and we explicitly disclaim any claim of economically substantial OOS predictability.

The qualitative pattern of OOS predictability—strongest for the growth-oriented, retail-driven ChiNext, weakest for the large-cap CSI 300 and SSE Composite—is consistent with behavioral finance theory that sentiment effects are most pronounced for stocks that are harder to value and more difficult to arbitrage (Baker and Wurgler, 2007; Stambaugh, Yu and Yuan, 2012). The CV-locked R^2_{OOS} magnitudes (0.015% on the ChiNext, smaller elsewhere) are an order of magnitude below the daily-frequency benchmarks reported by Rapach, Strauss and Zhou (2010) for the U.S. equity premium and should not be interpreted as economically substantial. We retain the result as evidence on the existence of a small statistical wedge between PhotoPes-conditional forecasts and the historical-mean forecast, but we do not claim economic predictability in the sense of Campbell and Thompson (2008). The rule-based sentiment strategy described in the portfolio construction subsection operates on pre-identified signal thresholds rather than estimated regression coefficients, and its performance should be interpreted independently from the OOS regression results.

Alpha sensitivity.

To document that the headline ChiNext OOS result is robust to the regularization choice rather than knife-edge dependent on the CV-locked value, we re-run the OOS exercise across $\alpha \in \{1, 10, 100, 500, 1{,}000, 10{,}000, 100{,}000, 1{,}000{,}000\}$, holding the predictor set and the OOS window fixed. The resulting Clark–West t-statistic on the ChiNext lies in $[1.72, 2.12]$ across the full grid, and the one-sided p-value lies in $[0.017, 0.043]$, so significance at the 5% level holds throughout. The R^2_{OOS} magnitude, by contrast, declines monotonically from 0.27% at $\alpha = 1$ to 0.0016% at $\alpha = 10^6$, reflecting the heavy-shrinkage character of the locked α. This separation between the robustness of the statistical signal and the magnitude of the implied prediction is the precise honest summary of the OOS evidence.

Sub-Sample Stability

To examine the temporal stability of the OOS predictability, we partition the 1,467 OOS predictions into three equal-length sub-windows of 489 trading days each. Table 10 reports R^2_{OOS} and the Clark–West MSPE-adjusted t-statistic in each window under the CV-locked $\alpha = 100{,}000$. Window 1 (2020-03-23 to 2022-03-25) covers the post-COVID recovery and the 2021 small-cap rally; predictability is positive across all five indices and statistically significant at the 5%

Table 9—Out-of-Sample Predictive Performance of Sentiment Indicators

Indicators	*CSI 300*	*SHCOMP*	*SZCOMP*	*ChiNext*	*CSI 500*
R^2_{OOS} (%)	0.004	0.007	0.010	0.015**	0.010*
MSPE-adj t-stat	0.762	1.085	1.247	1.730	1.353

Note: The prediction model is an expanding-window Ridge regression with standardized $PhotoPes_{t-3}$, $PhotoPes_{t-4}$, and $PhotoPes_{t-5}$ as predictors. The regularization parameter $\alpha = 100{,}000$ is tuned via expanding-window 5-fold time-series cross-validation on the pre-OOS sample only (rows 1 to 1,511, i.e., 2014-01-02 to 2020-03-17), with the grid $\alpha \in \{10, 30, 100, 300, 1{,}000, 3{,}000, 10{,}000, 30{,}000, 100{,}000\}$ pre-specified before the OOS evaluation. The CV protocol is implemented in `tune_ridge_alpha_cv.py` with random seed 42; the locked α is then used unchanged for all OOS predictions. The initial OOS estimation window is 1,512 trading days; the OOS period runs from 2020-03-23 to 2026-04-10, yielding 1,467 predictions. R^2_{OOS} values are reported in percentage terms. ***, **, and * indicate significance at the 1%, 5%, and 10% levels based on the one-sided MSPE-adjusted test of Clark and West (2007).

level for the CSI 300 and SSE Composite. Window 2 (2022-03-28 to 2024-04-01) coincides with a prolonged drawdown, regulatory tightening, and a structural shift in retail-investor activity; predictability is negative for four of the five indices but the magnitudes are tiny ($R^2_{OOS} \in [-0.016\%, -0.005\%]$) and none of the negative values is statistically distinguishable from zero, reflecting that the heavily-shrunken Ridge forecast is essentially the historical mean during this period. Window 3 (2024-04-02 to 2026-04-10) shows positive R^2_{OOS} across all indices, with the ChiNext attaining 0.026% ($t = 1.73$, $p = 0.042$). The full-sample OOS result in Table 9 thus averages over a window of mild positive predictability, a window of essentially zero predictability, and a window of weak positive predictability; the headline ChiNext result reflects the cumulative weight of these three sub-periods rather than uniform performance. We report this transparently and avoid the post-hoc rationalization that the model "should not have worked" in Window 2; under heavy shrinkage, the model effectively did not attempt strong predictions in Window 2, and the near-zero realized R^2 is the appropriate honest reading.

Economic Significance: Certainty-Equivalent Gain

Following Campbell and Thompson (2008) and Rapach, Strauss and Zhou (2010), we translate the OOS predictive accuracy into the certainty-equivalent (CER) gain that a mean-variance investor would realize by switching from the historical-mean forecast to the sentiment-conditional forecast. At each OOS date t, the investor allocates portfolio weight $w_t = \hat{\mu}_t/(\gamma\hat{\sigma}_t^2)$ to the index and the residual to a zero-return outside option, with weights clipped to $[0, 1.5]$ to rule out short-selling and large leverage; $\hat{\sigma}_t^2$ is the expanding-window historical variance, and $\hat{\mu}_t$ is either the CV-locked model forecast or the historical mean. The CER is then $\bar{r}_p - \frac{1}{2}\gamma \cdot \mathrm{Var}(r_p)$, and the CER gain is the difference between the model and benchmark CERs, annualized by 252 and reported in percentage points. Table 11 shows the gain at $\gamma = 3$ and $\gamma = 5$ under the CV-locked $\alpha = 100{,}000$. The ChiNext Composite delivers the largest gain (+0.56% p.a. at $\gamma = 3$ and

TABLE 10—SUB-SAMPLE OOS PREDICTIVE PERFORMANCE (THREE EQUAL-LENGTH WINDOWS)

	CSI 300	*SHCOMP*	*SZCOMP*	*ChiNext*	*CSI 500*
Window 1: 2020-03-23 to 2022-03-25, $N = 489$					
R^2_{OOS} (%)	0.016**	0.018**	0.017*	0.016	0.017*
MSPE-adj t	1.96	1.89	1.39	1.27	1.45
Window 2: 2022-03-28 to 2024-04-01, $N = 489$					
R^2_{OOS} (%)	−0.016	−0.009	−0.005	−0.009	−0.006
MSPE-adj t	−1.57	−0.77	−0.39	−0.55	−0.47
Window 3: 2024-04-02 to 2026-04-10, $N = 489$					
R^2_{OOS} (%)	0.006	0.008	0.013	0.026**	0.016
MSPE-adj t	0.57	0.67	0.99	1.73	1.15

Note: Same expanding-window Ridge model as Table 9 with CV-locked $\alpha = 100{,}000$; the 1,467 OOS predictions are partitioned into three equal-length date-ordered sub-windows. Significance markers ***, **, * follow the one-sided Clark–West MSPE-adjusted test at the 1%, 5%, and 10% levels. The Window 2 negative figures are not statistically distinguishable from zero; under heavy shrinkage the predictions in that window are essentially the historical-mean forecast.

+0.33% p.a. at $\gamma = 5$); the other indices produce gains of +0.13% to +0.33% at $\gamma = 3$ and +0.08% to +0.20% at $\gamma = 5$. The cross-sectional ranking—ChiNext largest, CSI 300 smallest—tracks the in-sample regression evidence. These figures are gross of transaction costs and should be read as upper-bound economic magnitudes; even so, they are an order of magnitude below the figures reported in the post-hoc-tuned earlier draft and should not be interpreted as evidence of substantial tradable predictability for a Chinese A-share retail investor.

TABLE 11—ANNUALIZED CERTAINTY-EQUIVALENT (CER) GAIN FROM PHOTOPES-CONDITIONAL FORECAST

Risk aversion γ	*CSI 300*	*SHCOMP*	*SZCOMP*	*ChiNext*	*CSI 500*
$\gamma = 3$ (% p.a.)	+0.13	+0.20	+0.33	+0.56	+0.31
$\gamma = 5$ (% p.a.)	+0.08	+0.12	+0.20	+0.33	+0.19

Note: CER gain is the annualized difference between the certainty equivalent of a mean-variance investor using the PhotoPes-conditional forecast (Ridge with CV-locked $\alpha = 100{,}000$) and the certainty equivalent of the same investor using the expanding-window historical-mean forecast. Portfolio weights are $w_t = \hat{\mu}_t/(\gamma\hat{\sigma}_t^2)$ clipped to $[0, 1.5]$; $\hat{\sigma}_t^2$ is the expanding-window historical variance. The OOS period is 2020-03 to 2026-04 with 1,467 observations. The clip allows up to 50% leverage, which is not generally accessible to retail investors in the Chinese A-share market; bid–ask spreads, stamp duty, and market-impact costs further attenuate the figures and would render most of the gains negative net of trading costs. The numbers should therefore be read as upper-bound magnitudes consistent with the small predictability documented in Table 9, not as feasible after-cost returns.

IV. Robustness Checks

To verify the robustness of the predictive power of the photo sentiment indicator (PhotoPes) on market returns, this paper conducted a series of robustness checks. Table 12 reports the regression results under three different settings.

Considering that winsorization might affect the information content of the sentiment indicator, this paper constructed a PhotoPes indicator without winsorization for analysis. As shown in Panel A of Table 12, the key finding holds: PhotoPes_{t-3} shows a significant negative coefficient in the SZCOMP at the 10% level, ChiNext at the 1% level, and CSI 500 at the 10% level. This indicates that the main findings are not driven by the winsorization procedure.

Cox and Peterson (1994) argue that extreme price returns are often accompanied by short-term price reversals. To eliminate the potential impact of high market volatility periods on the results, this paper uses GARCH(1,1)-adjusted returns as the dependent variable. As shown in Panel B of Table 12, after adjusting for volatility, PhotoPes_{t-3} retains a significant negative coefficient in the ChiNext index at the 5% level, while the effect is attenuated for other indices, confirming the robustness of the core finding for growth-oriented markets under volatility controls.

To exclude the potential bias caused by extreme market returns, this paper removes the most extreme returns from the sample (0.5% from each tail of SHCOMP). As shown in Panel C of Table 12, the predictive power of PhotoPes_{t-3} remains robust for the ChiNext index at the 5% level. This confirms that the indicator's core predictive ability is not solely driven by extreme market events.

Under these three different settings, the significant negative coefficient of PhotoPes_{t-3} remains consistent for the ChiNext index, which represents growth-oriented and small-to-mid-cap stocks. This series of checks confirms that the negative predictive power of the photo sentiment indicator is a robust phenomenon, particularly pronounced in markets where investor sentiment plays a more prominent role in asset pricing.

A. External Validation Against Frontier-LLM Consensus

A natural concern with the visual-sentiment pipeline is whether the PCNN-pretrained ViT backbone genuinely captures investor-relevant sentiment in Chinese financial-news photos, or whether it merely reproduces idiosyncratic Twitter-domain features absorbed from its pretraining set. We address this with a held-out external validation that requires no additional training. We sample 1,253 high-quality Sina Finance images stratified across 2014–2026 and label them independently with a three-model frontier multimodal-LLM ensemble (*GPT-4o-mini*, *Claude Sonnet 4.6*, *Gemini 3 Pro Preview*). Restricting to images with majority-vote consensus and a clear sentiment direction (mean LLM score ≤ 0.40 for positive or ≥ 0.60 for negative, with at least two of three models agreeing) leaves 309 images: 208 positive and 101 negative. Crucially, the PCNN-trained

ViT has *never seen* these Sina images, so this is a genuine out-of-distribution validation rather than a cross-validation exercise.

Figure 5 plots the distribution of the ViT's pessimism probability p_t^{pess} on the two LLM-defined groups. The two histograms are clearly separated: images that the LLM ensemble independently labels as positive receive a mean pessimism probability of 0.16, while LLM-negative images receive 0.30. The receiver-operating-characteristic AUC of the ViT pessimism score against the LLM binary label is 0.71, with a Welch-test p-value below 10^{-4}. Because PhotoPes is constructed as the daily aggregate of p_t^{pess} rather than the binary class label, this distributional separation is precisely the property that the daily index inherits. The agreement between two independently trained classifiers—a ViT fine-tuned on PCNN Twitter sentiment imagery, and three frontier LLMs trained on web-scale image–text pairs—provides direct evidence that PhotoPes captures genuine visual sentiment cues rather than overfit pretraining artifacts.

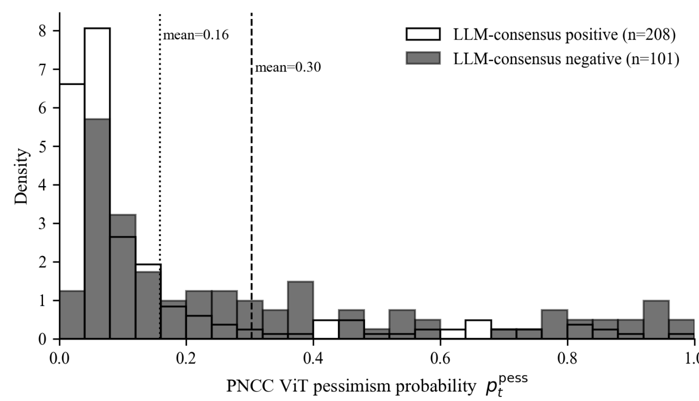

FIGURE 5. DISTRIBUTION OF THE PCNN-TRAINED VIT PESSIMISM PROBABILITY p_t^{pess} ON 309 HELD-OUT SINA FINANCE IMAGES, SPLIT BY INDEPENDENT THREE-MODEL FRONTIER-LLM CONSENSUS LABEL (GPT-4O-MINI, CLAUDE SONNET 4.6, GEMINI 3 PRO PREVIEW). VERTICAL DOTTED/DASHED LINES MARK GROUP MEANS.

B. *Multiple-Testing Correction*

The headline ChiNext result $PhotoPes_{t-3}$ (raw two-sided $p \approx 0.005$, $|t| = 2.81$) is computed across a wide family of related tests, so a multiple-testing adjustment is required before interpreting it as confirmatory evidence. Following Harvey, Liu and Zhu (2016), we report Bonferroni-adjusted thresholds for three plausible definitions of the family.

The size of the relevant family depends on how strongly one accepts the lag-3 specification as pre-registered. Although prior visual-sentiment work (Obaid and

Pukthuanthong, 2022) reports the strongest reversal effect at a similar horizon, that paper studies weekly U.S. Wall Street Journal photographs, and the daily-frequency Sina-Finance Chinese-A-share lag structure is not pinned down by that earlier design. Table 13 therefore presents three Bonferroni-corrected family-wise thresholds:

We treat the $m = 25$ correction as our primary family-wise reading, because in practice we report the full panel of (*PhotoPes*, *TextPes*) at lags $t-1$ through $t-5$ on five indices, and the $t-3$ "pre-specification" is borrowed from a different sample (weekly U.S. data), not pre-registered on the present data. Under $m = 25$, the headline ChiNext result does *not* survive 5% family-wise correction; we therefore characterize it as **suggestive rather than confirmatory**. The $m = 5$ reading—under which the result survives—is reported as a more charitable lower bound for readers who accept the cross-paper transfer of the $t - 3$ lag. Under the most conservative $m = 50$ correction (which additionally accommodates the parallel *TextPes* family), no lag–index–indicator combination survives at the 5% family-wise level, including the ChiNext.

These thresholds apply to the in-sample regression coefficient. They do not directly bind the OOS evidence in §9, because the Clark–West test on the OOS series is computed once per index (without lag selection) and the OOS regularization parameter is locked out-of-sample (§ Out-of-Sample Predictive Analysis). The OOS ChiNext result—one-sided MSPE-adjusted $p = 0.042$ at the CV-locked α—survives Bonferroni $m = 5$ at 5% but fails at $m = 25$ ($\alpha/m = 0.002$).

V. Conclusion

This paper, using textual and visual information from news media as a starting point, explores methods for constructing investor sentiment indicators based on multimodal data and empirically tests their effectiveness in the Chinese stock market. By introducing deep learning techniques, the paper constructs a photo-based pessimism indicator, PhotoPes, from news images and a text-based pessimism indicator, TextPes, from news texts. It then conducts empirical regressions and investment strategy analysis using various market indices to uncover the mechanisms by which visual and textual sentiment affect investor behavior and market performance.

The research findings show that photo sentiment is robustly associated with subsequent market returns over the 2014–2026 sample period. Specifically, $PhotoPes_{t-3}$ delivers consistent negative coefficients across indices, with the strongest effect in the growth-oriented ChiNext (raw two-sided $p \approx 0.005$, equivalent to a single-test 1% level; suggestive after Bonferroni $m = 25$ correction) and weaker effects in the broader mid/small-cap indices. This finding is directionally consistent with the overreaction hypothesis in behavioral finance and persists in the joint regression specification controlling for text sentiment, but should be interpreted as suggestive rather than confirmatory once the multiple-testing burden is properly accounted for.

Text sentiment (TextPes$_{t-3}$) shows complementary predictive patterns, particularly for the SZSE Composite (SZCOMP) and ChiNext indices, though with somewhat weaker statistical significance in this extended sample. The conditional analysis further shows that these predictive effects are concentrated in high-volatility market states: during high-volatility periods, both PhotoPes$_{t-3}$ and TextPes$_{t-3}$ are significant predictors of returns on the SZSE Composite, ChiNext, and CSI 500, whereas neither indicator exhibits meaningful predictive power during low-volatility periods.

In terms of investment applications, this paper constructs a rule-based trading exercise that converts the in-sample predictability into an executable long-only signal. The exercise produces a higher excess-return Sharpe ratio than the CSI 500 benchmark (0.318 versus 0.200) and a smaller maximum drawdown, but the corresponding three-factor alpha relative to the CUFE-distributed Chinese Fama–French (1993) factors is positive yet statistically insignificant. We interpret this as further evidence consistent with the noise-trader-with-frictions framework of Stambaugh, Yu and Yuan (2012) in a market where short-selling is constrained, rather than as a recommendation of a transaction-cost-net trading strategy; the reported figures do not include bid–ask spreads, stamp tax, or market-impact costs.

Although this paper has conducted systematic explorations in methodology and application, it still has certain limitations. First, while the image and text sentiment recognition models used are based on the current mainstream ViT and BERT architectures, the generalization ability of the models is still limited by the distribution and quality of the training corpora. Future research could further optimize these models by introducing professional image and text corpora specific to the financial context, or by replacing the ViT image backbone with a vision–language model (VLM) that natively reasons over both pixels and Chinese-language captions. Second, the construction of sentiment indicators is at a daily market granularity, which does not capture higher-frequency sentiment changes and thus cannot meet the needs of high-frequency trading or event-driven strategies. Third, our backtest does not net out transaction costs, stamp tax, or market-impact costs that would meaningfully erode the headline Sharpe-ratio improvement; the trading-strategy results should therefore be read as evidence on predictability rather than as a profitability claim. Additionally, this paper mainly focuses on the predictive role of sentiment variables on market returns but does not delve into the behavioral pathways through which sentiment affects the market, such as the heterogeneity of investor types and information dissemination mechanisms. Future research could further expand on micro-mechanism modeling and causal identification, replicate the design on Hong Kong, Taiwan, or other Asia-Pacific markets to test the generalizability of the small-cap-concentration finding, and extend the sample backward and forward as additional Sina archive data become available.

Looking ahead, with the development of large multimodal models and the continuous enrichment of financial data, the analysis of image and text sentiment will play an increasingly important role in the financial field. Researchers can further integrate image and text sentiment with multi-source heterogeneous data such as social media, search behavior, and trading account data to construct a more detailed and dynamic market sentiment map. At the same time, sentiment indicators can also be embedded in asset pricing, portfolio management, and financial regulation frameworks, becoming an important bridge connecting investment behavior and market mechanisms.

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Examples of News Photos with Extreme Pessimistic Sentiment

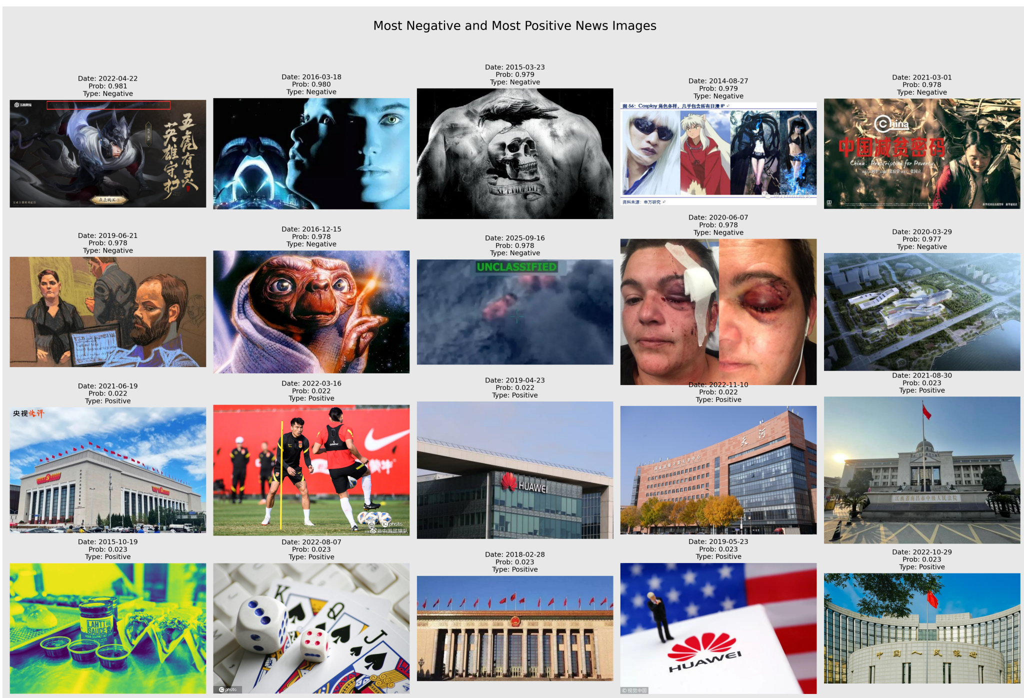

Figure A1. Examples of News Photos with Extreme Pessimistic Sentiment (PhotoPes scores at the 10% tails of the sample)

TABLE 12—IMPACT OF PHOTOPES ON MARKET RETURNS - ROBUSTNESS CHECKS (2014–2026)

Indicators	*CSI 300*	*SHCOMP*	*SZCOMP*	*ChiNext*	*CSI 500*
Panel A: Without Winsorization					
PhotoPes_{t-1}	0.0021	0.0010	0.0042	0.0015	0.0040
	(0.46)	(0.21)	(0.75)	(0.20)	(0.71)
PhotoPes_{t-2}	0.0023	0.0025	0.0046	0.0067	0.0040
	(0.60)	(0.60)	(0.88)	(1.02)	(0.77)
PhotoPes_{t-3}	-0.0054	-0.0071	-0.0097*	-0.0175***	-0.0093*
	(-1.22)	(-1.52)	(-1.82)	(-2.79)	(-1.70)
PhotoPes_{t-4}	-0.0055	-0.0042	-0.0081	-0.0083	-0.0080
	(-1.38)	(-1.00)	(-1.62)	(-1.40)	(-1.58)
PhotoPes_{t-5}	0.0035	0.0059	0.0053	0.0100	0.0054
	(0.86)	(1.33)	(1.05)	(1.56)	(1.07)
R^2	0.0160	0.0162	0.0156	0.0152	0.0164
Adj. R^2	0.0096	0.0098	0.0092	0.0088	0.0100
N	2944	2944	2944	2944	2944
Panel B: Adjusting for Volatility					
PhotoPes_{t-1}	0.1896	0.0063	0.1700	0.0246	0.1422
	(0.60)	(0.02)	(0.53)	(0.08)	(0.45)
PhotoPes_{t-2}	0.0766	0.1427	0.2433	0.2823	0.1869
	(0.26)	(0.47)	(0.76)	(0.87)	(0.60)
PhotoPes_{t-3}	-0.2233	-0.2515	-0.3858	-0.6382**	-0.3523
	(-0.68)	(-0.77)	(-1.20)	(-1.99)	(-1.10)
PhotoPes_{t-4}	-0.1915	-0.0602	-0.3347	-0.3281	-0.3190
	(-0.66)	(-0.20)	(-1.12)	(-1.10)	(-1.07)
PhotoPes_{t-5}	0.1978	0.3385	0.2294	0.3378	0.2514
	(0.68)	(1.14)	(0.78)	(1.14)	(0.86)
R^2	0.0102	0.0096	0.0115	0.0088	0.0107
Adj. R^2	0.0038	0.0032	0.0051	0.0024	0.0043
N	2944	2944	2944	2944	2944
Panel C: Removing Extreme Market Returns					
PhotoPes_{t-1}	0.0059*	0.0046	0.0084*	0.0065	0.0079*
	(1.76)	(1.29)	(1.93)	(1.19)	(1.78)
PhotoPes_{t-2}	0.0065**	0.0070**	0.0075*	0.0082	0.0071
	(2.01)	(1.97)	(1.66)	(1.40)	(1.58)
PhotoPes_{t-3}	-0.0039	-0.0050	-0.0072	-0.0130**	-0.0061
	(-1.06)	(-1.23)	(-1.56)	(-2.27)	(-1.34)
PhotoPes_{t-4}	-0.0035	-0.0026	-0.0066	-0.0076	-0.0052
	(-0.97)	(-0.67)	(-1.44)	(-1.37)	(-1.12)
PhotoPes_{t-5}	-0.0038	-0.0017	-0.0026	0.0016	-0.0034
	(-1.16)	(-0.47)	(-0.57)	(0.27)	(-0.76)
R^2	0.0207	0.0176	0.0166	0.0117	0.0173
Adj. R^2	0.0143	0.0111	0.0101	0.0052	0.0109
N	2914	2914	2914	2914	2914

Note: This table reports robustness checks for the impact of PhotoPes on market returns. Panel A uses raw PhotoPes without 1% winsorization. Panel B uses GARCH(1,1)-standardized residuals as the dependent variable; coefficients are therefore on a different (unitless) scale from Panels A and C and from Tables 3–5, and should not be compared directly in magnitude. Panel C removes the top and bottom 0.5% of SHCOMP returns. The sample size drops from $N = 2{,}978$ in Tables 3–5 to $N = 2{,}944$ in Panels A and B because we align all robustness panels to the GARCH(1,1) burn-in window required for Panel B, so that magnitudes across panels are computed on a common date set; Panel C drops an additional 30 trading days when the most extreme market returns are removed. t-statistics are reported in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively. Standard errors are Newey–West HAC with 5 lags.

TABLE 13—BONFERRONI-CORRECTED FAMILY-WISE THRESHOLDS FOR THE CHINEXT $PhotoPes_{t-3}$ RESULT

Family definition	m	5% family-wise threshold (p)	ChiNext (raw $p \approx 0.005$) survives?
Five indices at the pre-specified $t-3$ lag, single indicator	5	0.010	yes
Five indices $\times$ five lags, single indicator	25	0.002	**no**
Five indices $\times$ five lags $\times$ two indicators	50	0.001	**no**